Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Sensory Pathways — Vision & Hearing

MCB80x · receptor → brainstem → thalamus → cortex · visual field lesions

VISUAL PATHWAY — EYE TO CORTEX

Receptor · Nerve / tract · Brainstem · Thalamus · Cortex

Retina	Optic nerve	Optic chiasm	Optic tract	LGN	Optic radiations	V1 → streams
Eye Rods (dim light) and cones (color) → bipolar → ganglion cells.	Cranial nerve II Ganglion cell axons leave the eye; blind spot where they exit.	Below hypothalamus **Nasal** fibers cross; temporal fibers stay on the same side.	Each carries one visual field Right tract = left visual field from both eyes.	Thalamus 6 layers; keeps eyes and M/P streams separate.	White matter Fan out through parietal and temporal lobes to V1.	Occipital lobe **Ventral** "what" → temporal. **Dorsal** "where/how" → parietal.

AUDITORY PATHWAY — EAR TO CORTEX

Cochlea	Auditory nerve	Cochlear nuclei	Superior olive	Inferior colliculus	MGN	A1
Inner ear Hair cells bend → transduce sound. Basilar membrane maps frequency.	Cranial nerve VIII Carries frequency (place) and timing information.	Medulla First central stop; still one ear only.	Pons **First binaural** stop: compares the two ears to localize sound.	Midbrain Integrates all brainstem auditory input.	Thalamus Medial geniculate relays to cortex.	Temporal lobe Tonotopic map on Heschl's gyrus.

VISUAL FIELD LESIONS (black = lost field)

Lesion site	L eye · R eye	Deficit
Right optic nerve		Blind in the right eye.
Optic chiasm (center)		**Bitemporal hemianopia**, often from a pituitary tumor.
Right optic tract		**Left homonymous hemianopia** (same half lost in both eyes).
Right V1		Left homonymous hemianopia, often with **macular sparing**.

PRINCIPLES TO KNOW

- **Thalamic relay:** vision (LGN) and hearing (MGN) both pass the thalamus; only smell skips it.
- **Maps are kept:** retinotopy in vision, tonotopy in hearing, all the way to cortex.
- **Vision crosses at the chiasm**, so each hemisphere sees the opposite visual field.
- **Hearing is bilateral above the cochlear nuclei**, so one-sided cortex damage rarely deafens an ear.
- **Superior colliculus** (midbrain) also gets retinal input for reflexive eye and head orienting.

Personal MCB80x study chart — not an official HarvardX/edX publication.